

NSDUH Analysis Document

Isaac Moua

May 14th, 2026

This analysis project explores possible patterns of mental health treatment using data analyzed using the NSDUH database provided from the SAMHSA. The central question that guides this project is: "How do demographics and socioeconomic factors such as age, income and insurance coverage affect the likelihood of receiving mental health treatment?" By utilizing structured datasets with visualizations and interactive tools, we can attempt to identify disparities in access to care and identify trends that could provide useful information to positively change access to treatment..

Initial findings from analyzing the data revealed noticeable patterns. To start, there is a suggestion that younger age groups in adolescents, as well as middle-aged age groups have lower rates of treatment access than the rest. This result can be an indication of possible barriers in access to health care based on other correlating factors that are directly impacted from these filtered age groups. Additionally, other possible factors outside the scope of the dataset can be applied here, such as lack of awareness or social stigma, however due to the limitations of the application we cannot dive into this possibility. We also have to take in the factor of middle-aged groups being affected alongside younger age groups, as this statement may be affected by other factors instead of age relativity such as economics.

I have also found correlations of income distribution and its large role in influencing treatment accessibility. The data indicated a relationship between income and the likelihood of treatment, and is further strengthened with its connection when comparing reports of the treatment barrier "can not afford" with the survey income reports, as this barrier and specific group are the highest count-wise from the dataset. Another aspect of economical coverage to look over is insurance, however when we looked into the statistics we noted that there is a very large portion of insured reports that had barrier reports such as "Insurance did not cover price."

This statement can further push the conclusion of income being the highlight factor as we can conclude that insurance does not have a major role in treatment rates..

While these findings provide valuable insight, several limitations must be considered. First, the dataset used in this analysis is based on survey data, which may include self-reporting bias and inaccuracies. As for other limitations to the application and database we used, the dataset is specifically filtered using age, income, insurance and treatment. This choice was due to the extensive information within the dataset, which due to scalability and timing, I decided to cut down the variables we will be focusing on for simplified and generalized data. Due to this restriction, we disregard other possible factors that may influence our findings for health care access. While keeping the project on the smaller scale there is also a possibility in lack of UI/UX implementation, while we are able to view treatment in comparison to different key components, we then restrict our comparison models to only these aspects. Another limitation is that the analysis is primarily observational and does not establish causation—only correlations between variables.

In summary, this project's analysis of the NSDUH dataset concluded that there were strong association of age and income when it comes to mental health treatment rates. The findings highlight disparities in access to care among both younger adolescents and middle-age individuals, as well as lower income. However, we also know that insurance status does not guarantee treatment access for those who are capable income wise, this is seen with a large amount of reports of lack of coverage regardless of insurance. While this is our conclusion for the project, future work could expand on this analysis by incorporating predictive modeling, geographic data, and better usage of the provided database to better understand causal relationships and improve mental health interventions.